Project 4: SR Latch

Transistors are the foundational building blocks for how many complex circuits work today, being the building blocks for digital logic and memory/saved states. This project will help you to understand how NMOS transistors can be used to implement digital logic circuits.

Objectives

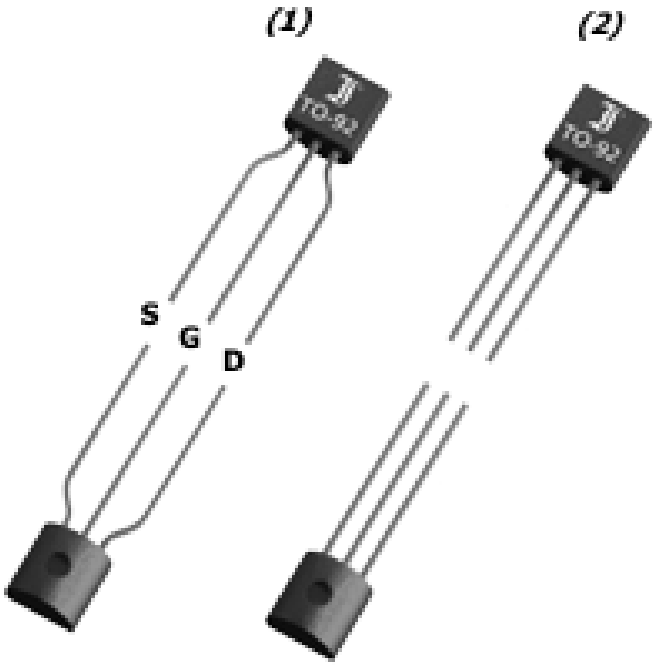
- Demonstrate how logic gates (**NOR**) are created using NMOS devices
- Build and test a functional **SR Latch** (set-reset latch) using NMOS transistors
- Use push buttons to provide input
- Observe and verify the latch's state-holding behavior

Task Information

What you will need:

- 1x Arduino Nano (and Power Cord)
  - https://a.co/d/0c8U6o9C
- 2x LED
  - https://www.digikey.com/en/products/detail/qt-brightek-qtb/QBL8IW15C-CW-2897/14664272
- 2x Pushbutton
  - https://www.digikey.com/en/products/detail/cui-devices/TS02-66-55-BK-160-LCR-D/15634374
- 1x Breadboard
- 1x Wire Kit
- 6x 330 Ohm Resistors
  - https://www.digikey.com/en/products/detail/stackpole-electronics-inc/CF14JT330R/1741399
- 4x 2N7000 NMOS transistors
  - https://www.digikey.com/en/products/detail/diotec-semiconductor/2N7000/13164314

NMOS Pinout:



DISCLAIMER

Because of the nature of resistor-load NMOS circuits, the current that travels from VDD to the output LED is lower than ideal due to the fact that the resistor decreases the current coming from the supply rail. Because of this, your LEDs will appear dim. I recommend using red LEDs as those will be most visible. If your LEDs are dim, it is most likely not an error; just a result of the type of circuits we are building.

This happens because the pull-up resistor and LED are in series between 5V and ground, forming a voltage divider that limits available current to the LED. A PMOS transistor used as an active load instead of a resistor would provide stronger drive current, which is how modern CMOS logic gates are built. If you want to learn more, message Lathan!

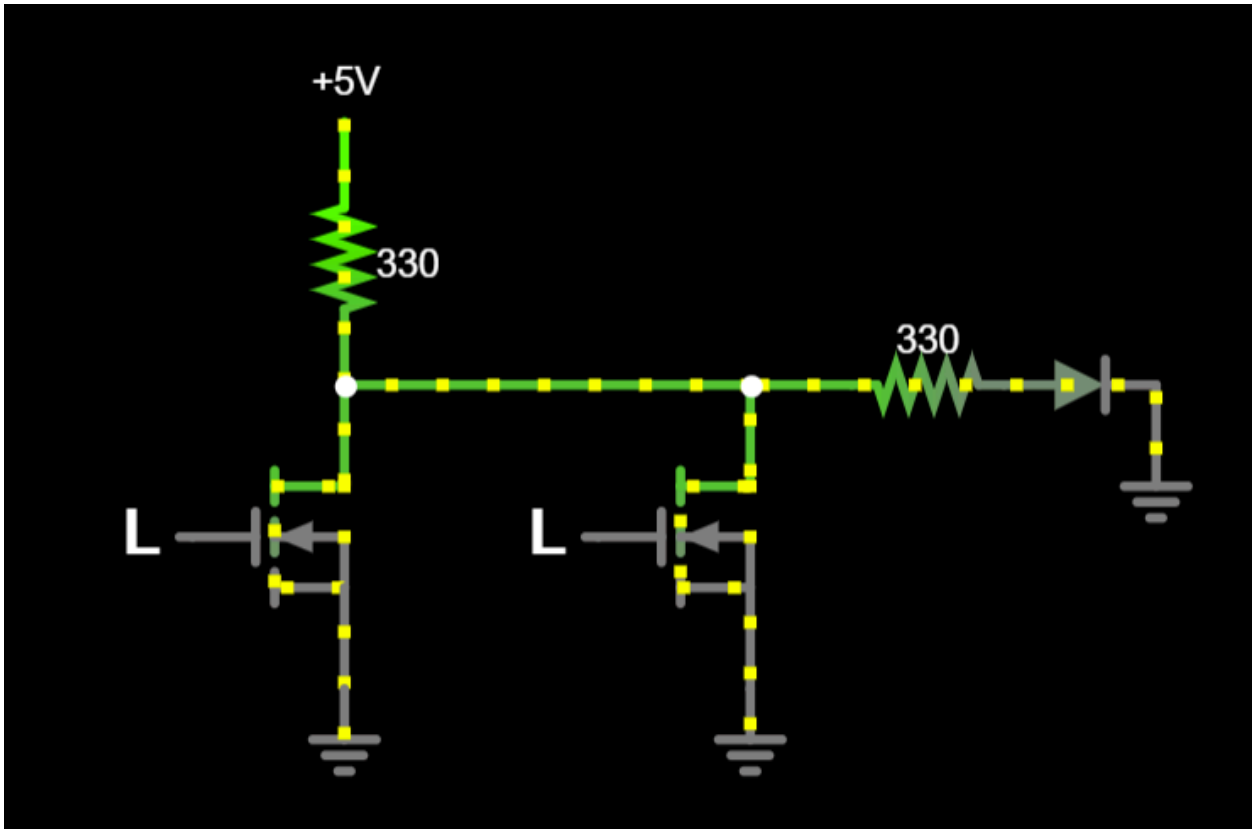
Part 1: NOR Gate

In Part 1 of this project, you will design a NOR gate using NMOS transistors and 330 Ohm load resistors.

2-input NOR gate truth table:

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

NOR gate transistor-level schematic:



This image shows the transistor-level schematic of a NOR gate with 330 Ohm load resistors. Here, 'L' represents logic low or '0'. 'H' represents logic high or '1'. However, in your circuit, you will wire the gates of the NMOS transistors to two buttons to serve as inputs.

NOR gate transistor-level simulation walkthrough: https://youtu.be/rAIB4UibOJg

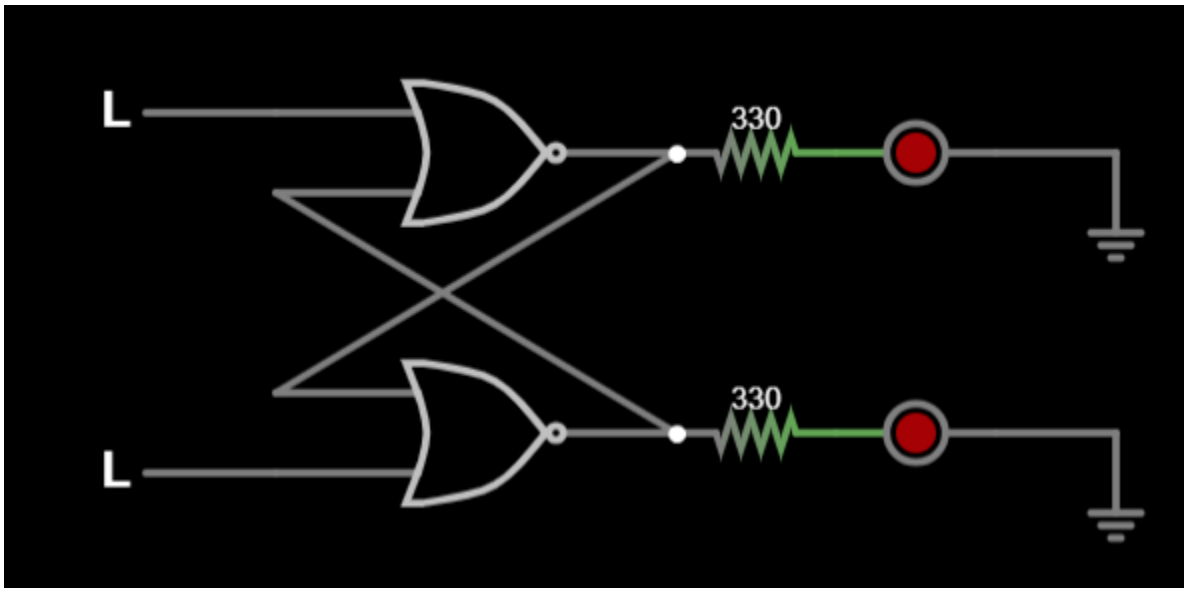
Part 2: SR Latch

Hint: I recommend that you create two NOR circuits on your breadboard before going straight into the SR Latch. Keep the two NOR circuits separated enough so that you won't have as many debugging headaches when you wire the NORs together. When you have two working NOR gates, it will be easier for you to identify what you need to change to connect the NORs together to make an SR Latch.

SR Latch truth table:

S	R	Q	Q'	State
0	0	Q	Q'	Hold
0	1	0	1	Reset
1	0	1	0	Set
1	1	X	X	Invalid

SR Latch gate-level schematic:



Like the NOR gate, here, 'L' represents logic low or '0'. 'H' represents logic high or '1'. However, in your circuit, you will wire the inputs shown in this schematic to two buttons to serve as your 'Set' and 'Reset' inputs.

SR Latch gate-level simulation walkthrough: https://youtu.be/svKlDl7sCQg

What to Submit

Due Date: Dec 1, 2026

Submit your completed **Part 2** circuit demonstration to the **Project 4 assignment on UF Check-In** as an **unlisted YouTube video link**. If you completed your project in office hours, be sure to demo your SR Latch to a coordinator, and then you can complete the assignment on UF Check-In by submitting "Completed in OH" rather than submitting your YouTube link. **You do not need to submit Part 1.** Remember, if you ever have trouble with OPS, we are here to help! Feel free to reach out on the Slack discussion channel, contact OPS coordinators directly, or attend office hours.

Resources

Lecture slides:
https://github.com/IEEE-University-of-Florida/ops-resources-26-27/tree/main/L4%20-%20Digital%20Logic%20and%20Transistors

Transistors video by Lathan (UF Silicon Design Team):
https://www.youtube.com/watch?v=HU8MGdQO-JU

Digital Logic video by Lathan (UF Silicon Design Team):
https://www.youtube.com/watch?v=bNKhy2vlivo

Transistor Applications video by Lathan (UF Silicon Design Team):
https://www.youtube.com/watch?v=8WGISh4FITs

Falstad Circuit Simulator:
https://www.falstad.com/circuit/circuitjs.html

Hints:

- These circuits can be complicated. Building slowly, methodically, and neatly is essential for success.
- Be sure you are double-checking your breadboard against the schematics to ensure that you are connecting components correctly, especially when you move on to the SR Latch.
- If you are building a push button circuit, make sure to add the appropriate pull up or pull down resistor to provide a default value when not pressed. Floating voltages are a large source of unpredictable behavior for digital circuits.